



Lecturer: Nadim Kobeissi

Website: <https://appliedcryptography.page>

Quiz 1.4: Pseudorandomness

This completely optional quick quiz acts as a learning aid to help you find out how much you absorbed from our Pseudorandomness session.^a Remember, this is just for fun and to help you identify areas you might want to review. Each question has three possible answers, but only one is correct. Take your time, and when you're done, check your answers against the answer key at the end. Good luck!

^a<https://appliedcryptography.page/slides/#1-4>

Questions

1. What is the main limitation of the one-time pad encryption scheme?
 - A. The key must be as long as the plaintext
 - B. It is computationally expensive
 - C. It doesn't provide perfect secrecy
2. What is a pseudorandom generator (PRG)?
 - A. A function that generates truly random numbers
 - B. A function that expands a short seed into a longer output indistinguishable from random
 - C. A function that compresses data
3. What is the primary difference between a PRG and a PRF?
 - A. PRFs are faster to compute than PRGs
 - B. PRGs are more secure than PRFs
 - C. PRFs provide selective access to pseudorandom values for any input
4. Which real-world cryptographic primitive is an example of a PRF?
 - A. Hash functions like SHA-256
 - B. Stream ciphers like ChaCha20
 - C. Block ciphers like AES
5. What property must a hash function have to be considered secure?
 - A. It must produce variable-length outputs
 - B. It must be collision resistant
 - C. It must be reversible
6. What is a pseudorandom permutation (PRP)?
 - A. A keyed bijective function indistinguishable from a random permutation
 - B. A one-way function that cannot be inverted
 - C. A function that randomly shuffles data
7. What is the key advantage of PRPs over PRFs?
 - A. PRPs are faster to compute

- B. PRPs provide better security
 - C. PRPs are efficiently invertible
8. How many rounds does a Feistel cipher need to be a secure PRP?
- A. 2 rounds
 - B. 3 or more rounds
 - C. 16 rounds
9. What makes Feistel ciphers always permutations?
- A. They require reversible round functions
 - B. They use strong encryption algorithms
 - C. Each round is invertible without inverting the round function
10. What is the main weakness of RC4?
- A. Statistical biases in its output
 - B. It is too slow for practical use
 - C. It has a small key size
11. What is AES an example of?
- A. A pseudorandom permutation
 - B. A pseudorandom generator
 - C. A hash function
12. What attack can break a 2-round Feistel cipher?
- A. Birthday attack on the round function
 - B. Observing that XOR of outputs equals XOR of inputs when using same right half
 - C. Exhaustive key search
13. What is the “Golden Rule of PRFs”?
- A. PRFs should use large key sizes
 - B. Security depends on ensuring distinct inputs to the PRF
 - C. PRFs must be implemented in hardware
14. What was the purpose of the EFF’s “Deep Crack” machine?
- A. To break AES encryption
 - B. To develop a new encryption standard
 - C. To demonstrate that 56-bit DES keys were insufficient
15. What is the difference between a PRF and a PRP in terms of output collisions?
- A. Both have the same collision probability
 - B. PRFs never have collisions
 - C. PRPs never have collisions, PRFs can have collisions
16. What is the main security concern with using the same key for all rounds in a Feistel cipher?
- A. It reduces the block size
 - B. It makes encryption slower
 - C. It enables meet-in-the-middle attacks
17. What technique was used to prove the security of 3-round Feistel ciphers?
- A. Linear cryptanalysis
 - B. Differential cryptanalysis
 - C. Bad event proof technique

18. What is the best known attack complexity against AES-128?
- A. Approximately 2^{126} operations
 - B. 2^{64} operations
 - C. 2^{256} operations
19. What does 3DES use to increase security over regular DES?
- A. More Feistel rounds
 - B. Larger block size
 - C. Three DES operations in sequence
20. What property makes something “pseudorandom” in cryptography?
- A. It is computationally indistinguishable from uniform distribution
 - B. It passes statistical randomness tests
 - C. It uses a deterministic algorithm

Answers

Question	Key
1	A
2	B
3	C
4	A
5	B
6	A
7	C
8	B
9	C
10	A
11	A
12	B
13	B
14	C
15	C
16	C
17	C
18	A
19	C
20	A